

Project Report

Introduction:

Coronary Artery Disease (CAD) is one of the leading causes of morbidity and mortality worldwide, making early detection and accurate diagnosis crucial for effective treatment. X-ray Coronary Angiography (XCA) is a widely used imaging technique that allows clinicians to visualize the coronary arteries and assess for abnormalities such as stenosis or plaque buildup, which are indicative of CAD. The ARCADE (Automatic Region-based Coronary Artery Disease Diagnostics using X-ray Angiography Images) dataset offers a rich collection of XCA images designed to facilitate the development of automated diagnostic tools for CAD. In this project, we explore how semantic segmentation can be applied to XCA images to detect coronary vessel tree.

Problem Statement:

The goal of this project is to develop a deep learning model for semantic segmentation of coronary vessel tree. The model aims to identify and segment coronary vessel tree from XCA images. Accurate segmentation of these regions enables identification of coronary arteries, simplifying the diagnostic process and contributing to the early detection of CAD. In this project, rather than segmenting coronary artery into different categories, we generate binary masks for coronary vessel tree in all the images and would not be dividing them into different regions/categories.

Datasets:

For this project, we use the ARCADE (Automatic Region-based Coronary Artery Disease Diagnostics using X-ray Angiography Images) dataset [1], which provides a collection of X-ray Coronary Angiography (XCA) images aimed at supporting the development of automated diagnostic systems for Coronary Artery Disease (CAD). The ARCADE dataset includes a total of 1200 coronary vessel tree images, with corresponding annotations that divide the heart into 26 distinct regions. These annotations are valuable for a more granular understanding of CAD, but for the scope of this project, we focus on generating binary masks to represent the entire coronary vessel tree without segmenting the individual regions.

The dataset is split into two groups:

- 1000 training images
- 200 validation images

Below are examples of sample image along with their corresponding combined segmentation masks and overlay on original image from the dataset:

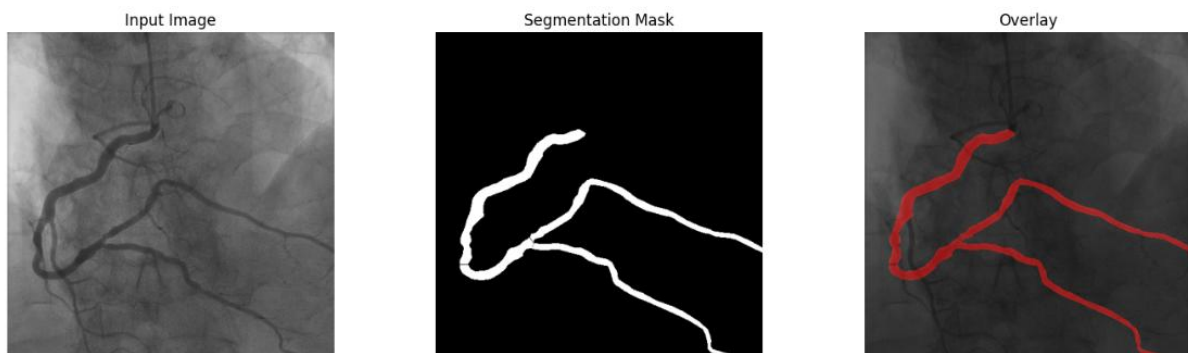


Figure 1.1 Dataset sample – Input image along with corresponding annotated mask and overlay

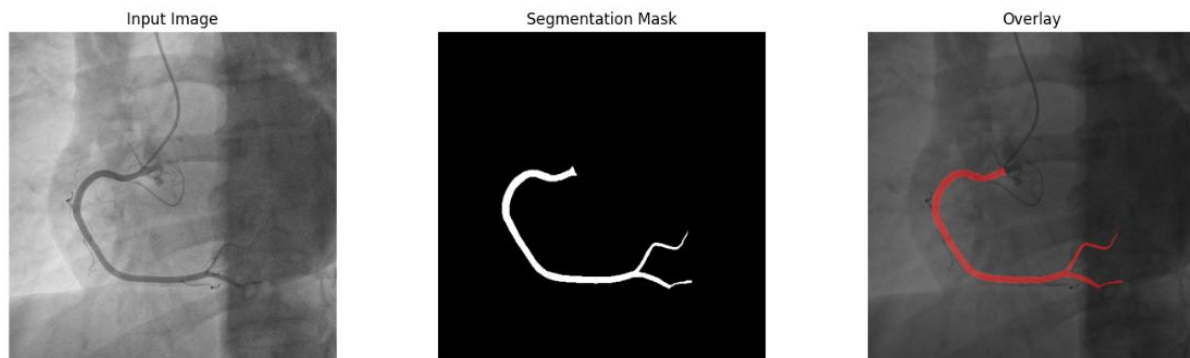


Figure 1.2: Dataset sample – Input image along with corresponding annotated mask and overlay

Model Architecture:

The model used in this project is based on the U-Net [2] architecture, which is widely used for semantic segmentation tasks. U-Net is a fully convolutional network designed for image segmentation, particularly in medical imaging, due to its efficiency in handling high-resolution images. The architecture follows an encoder-decoder structure with skip connections, allowing the model to retain spatial information during the down-sampling and up-sampling processes. This structure is particularly useful for segmenting intricate structures, such as the coronary vessel tree in XCA image

Adapted Network Components:

- **DoubleConv Layer:**
This block consists of two convolutional layers, each followed by batch normalization and a ReLU activation function. It is used to extract features from the input image and is repeated throughout both the encoder and decoder stages of the network.
- **Down Layer (Max Pooling + DoubleConv):**
The down-sampling blocks consist of a max pooling layer followed by a DoubleConv block. This allows the model to reduce the spatial dimensions of the image while learning more abstract features at each level.
- **Up Layer (Transposed Convolution + DoubleConv):**
The up-sampling layers use transposed convolutions to increase the spatial resolution of the feature maps. After each up-sampling, skip connections are employed, where features from the corresponding down-sampling stage are concatenated to help preserve spatial details.
- **Output Convolution:**
The final output layer produces a single channel tensor, representing the binary mask of the coronary vessel tree. This project focuses on binary segmentation, with the output representing the coronary arteries and their abnormalities as either foreground (1) or background (0).
- **Full Model (U-Net)**
The full U-Net model consists of the following components:
 - Encoder: A series of down-sampling layers (DoubleConv + MaxPool2d), which progressively reduce the spatial dimensions of the input while learning abstract feature representations.
 - Decoder: A series of up-sampling layers (ConvTranspose2d + DoubleConv), which progressively recover the original spatial dimensions. Skip connections from the encoder are used to provide high-resolution feature maps at each decoder stage.
 - Output Layer: A final convolutional layer that reduces the output to a binary mask representing the segmented region.

Training Procedure:

The model was trained on input images of shape (512, 512, 3). The model was trained using two different loss functions: Binary Cross Entropy Loss (BCE Loss) and Dice Loss [3], which is designed to address class imbalance. Each of these were trained using 2 different types of optimizers, i.e. SGD and ADAM. The hyperparameter values are listed below:

Optimizer	Learning Rate	Weight decay	Momentum
Adam	1e-4	1e-5	-
SGD	1e-2	1e-4	0.9

Dice Loss was used to address the class imbalance often encountered in segmentation tasks, where the foreground (target) occupies a small portion of the image compared to the background. Dice Loss directly optimizes the overlap between predicted and ground truth masks, making it more effective for small object segmentation. Unlike BCE, which can favor the background due to its pixel dominance, Dice Loss focuses on maximizing the intersection between the predicted and true regions of interest. This helps the model better segment smaller structures, like vessels, ensuring improved accuracy even when the target class is significantly underrepresented.

All the experiments were run for 20 epochs. Manual seed was set for reproducibility of results. The scheduler is used to adjust the learning rate during training based on the validation loss. The “ReduceLROnPlateau” scheduler decreases the learning rate by a factor of 0.5 if the validation loss does not improve for 3 consecutive epoch. The Mean Intersection over Union (IoU) metric is used for evaluating the model’s performance in terms of pixel-wise accuracy. The IoU metric computes the overlap between the predicted and true segmentation masks. The best model is saved based on validation loss, and visualizations of predictions are generated periodically.

Results and Evaluation

To evaluate the performance of the segmentation model, Mean Intersection over Union (IoU) metric was used. The IoU metric computes the overlap between the predicted and true segmentation masks.

For all the 4 experiments, the training and validation loss were monitored across epochs to evaluate the model's convergence and generalization. The learning curves for each loss function along with the prediction samples are shown below and help visualize training stability and outputs.

Using SGD optimizer:

- For BCE loss:

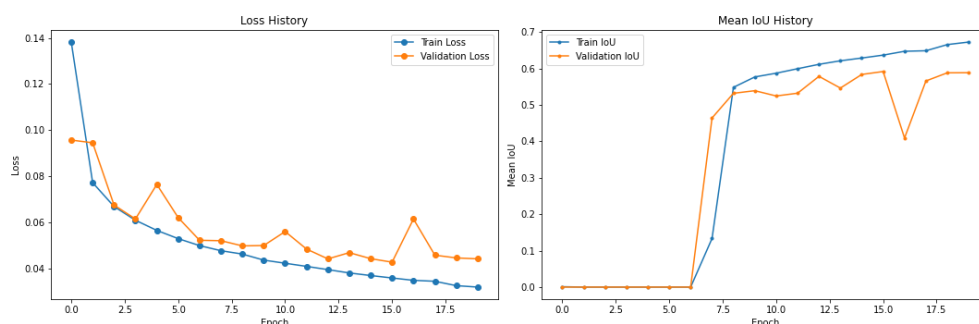


Figure 2.1 Model performance over epochs: loss (left) and mean IoU (right)

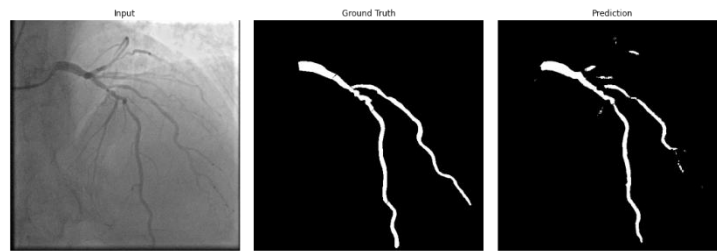


Figure 2.2: Model prediction

- For Dice loss:

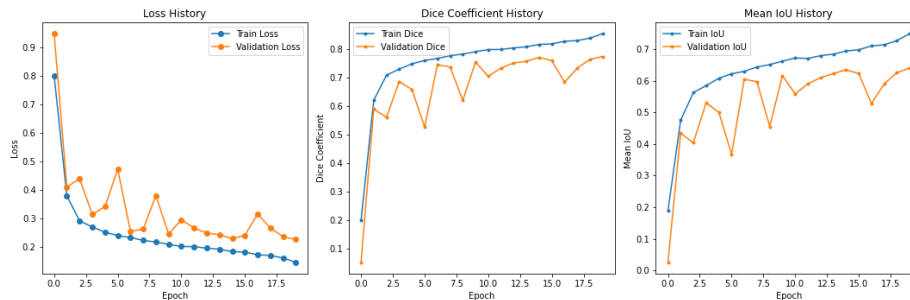


Figure 2.3: Metrics over epochs: Dice loss, Dice coefficient, and mean IoU

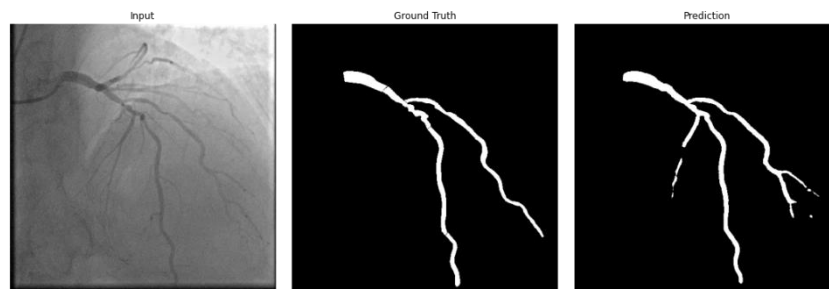


Figure 2.4: Model Prediction

- The model trained using the Dice loss along with SGD optimizer performs better than the model trained using BCE loss with SGD optimizer. The prediction from dice loss has more detail and predicts better mask when compared to BCE loss. BCE loss prediction is able to predict the overall mask but has more broken segments in prediction of the coronary vessels compared to Diceloss

Using Adam optimizer:

- For BCE loss:

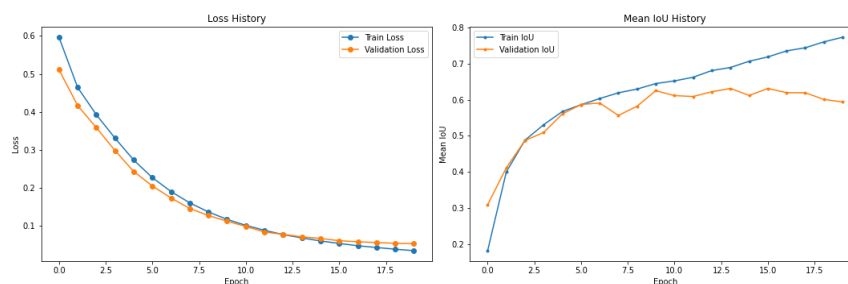


Figure 3.1: Model performance over epochs: loss (left) and mean IoU (right)

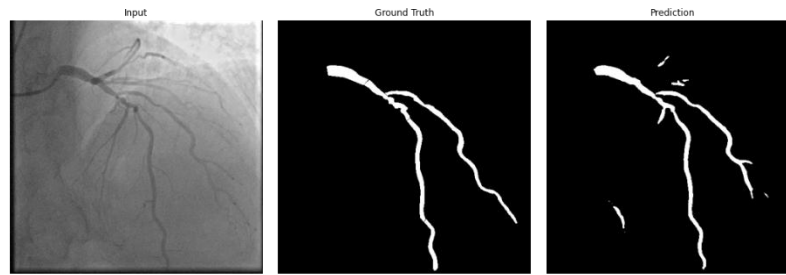


Figure 3.2: Model Prediction

- For Dice loss:

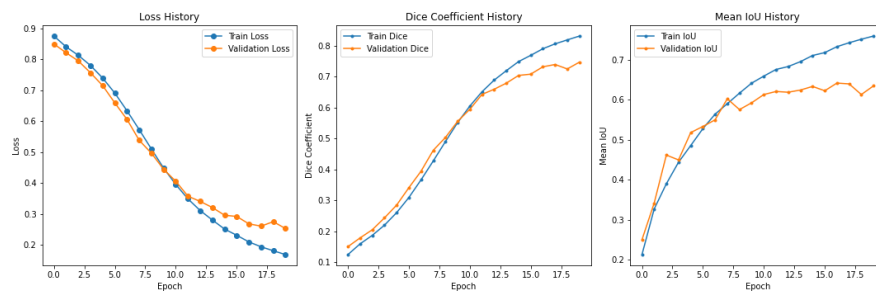


Figure 3.3: Metrics over epochs: Dice loss, Dice coefficient, and mean IoU

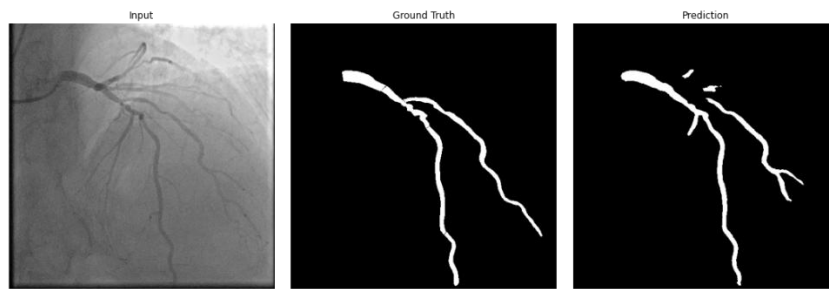


Figure 3.4: Model Prediction

- The model trained using the Dice loss along with Adam optimizer performs almost slightly better than model trained using BCE loss with Adam optimizer. The prediction from BCE loss has detected some more false positives as mask when compared to the model trained with Dice loss.
- Overall, we see that Dixeloss performs better than the BCEloss.
- Comparison of results of all 4 experiments are shown below:

Loss function	Optimizer	Train mIoU	Val mIoU
BCE loss	SGD	0.64	0.59
Dice loss	SGD	0.75	0.64
BCE loss	Adam	0.77	0.63
Dice loss	Adam	0.76	0.64

Future Work

As mentioned earlier, in this work, we only focused on generating a binary mask of the coronary tree vessels. For the next steps:

- We can extend the model to multiclass segmentation (e.g., separate segmentation of different vessel types present in data) which could provide more detailed insights into the vascular structures, which would be useful for more advanced diagnostic tools, for example stenosis.

- We can use different data augmentation techniques and see how they impact the overall mean IoU.
- We can try different segmentation model architecture and compare their performance from the existing ones
- Even though the input images contain three channels, they are fundamentally grayscale. In the next phase, the model could be retrained using images with a single grayscale channel instead of three. This change might lower computational costs and could enhance model performance by eliminating redundant information

References:

- [1] Dataset, "ARCADE", (Automatic Region-based Coronary Artery Disease Diagnostics using X-ray Angiography Images). Link: <https://zenodo.org/records/8386059>
- [2] O. Ronneberger, P. Fischer, and T. Brox, 'U-net: Convolutional networks for biomedical image segmentation', in *Medical image computing and computer-assisted intervention--MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18, 2015, pp. 234-241.*
- [3] Sudre, C., et al, "Generalised dice overlap as a deep learning loss function for highly unbalanced segmentations," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support: Third International Workshop, DLMIA 2017, and 7th International Workshop, ML-CDS 2017, Held in Conjunction with MICCAI 2017, Quebec City, QC, Canada, September 14, Proceedings 3, 2017, pp. 240-248.*